Carnegie Mellon University
Computer Science Academy

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

While preparing for the midterm exams, I have envisioned something I want to do. I want to create a "Kahoot"-like page that asks questions to the user. I also want to ensure that I have a section that tracks the scores, a question bar, and a question number. This will be simple, and I may make around 10 questions related to what I have learned in Computer Science Foundations. After answering all the questions correctly, I want a block to pop up showing that the user has successfully finished the quiz along with their score. On the other hand, maybe a block can show up saying that the user has got the question wrong; along with this, the program will stop. It would be great if I had 4 possible answers for each question or even typed-up answers, but I know this will not be possible.
A starting page is also applicable, which includes a start button and a short title for the quiz, just like Kahoot. One major thing I want to overcome is timing. During the second day of programming, my teacher suggested adding a timer, as online quizzes are usually timed. This is a great idea, and I will try to do this with the help of AI, as I have never learned how to do this.

Canvas Design

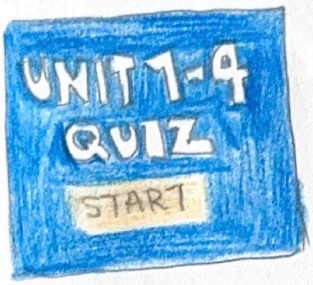
Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

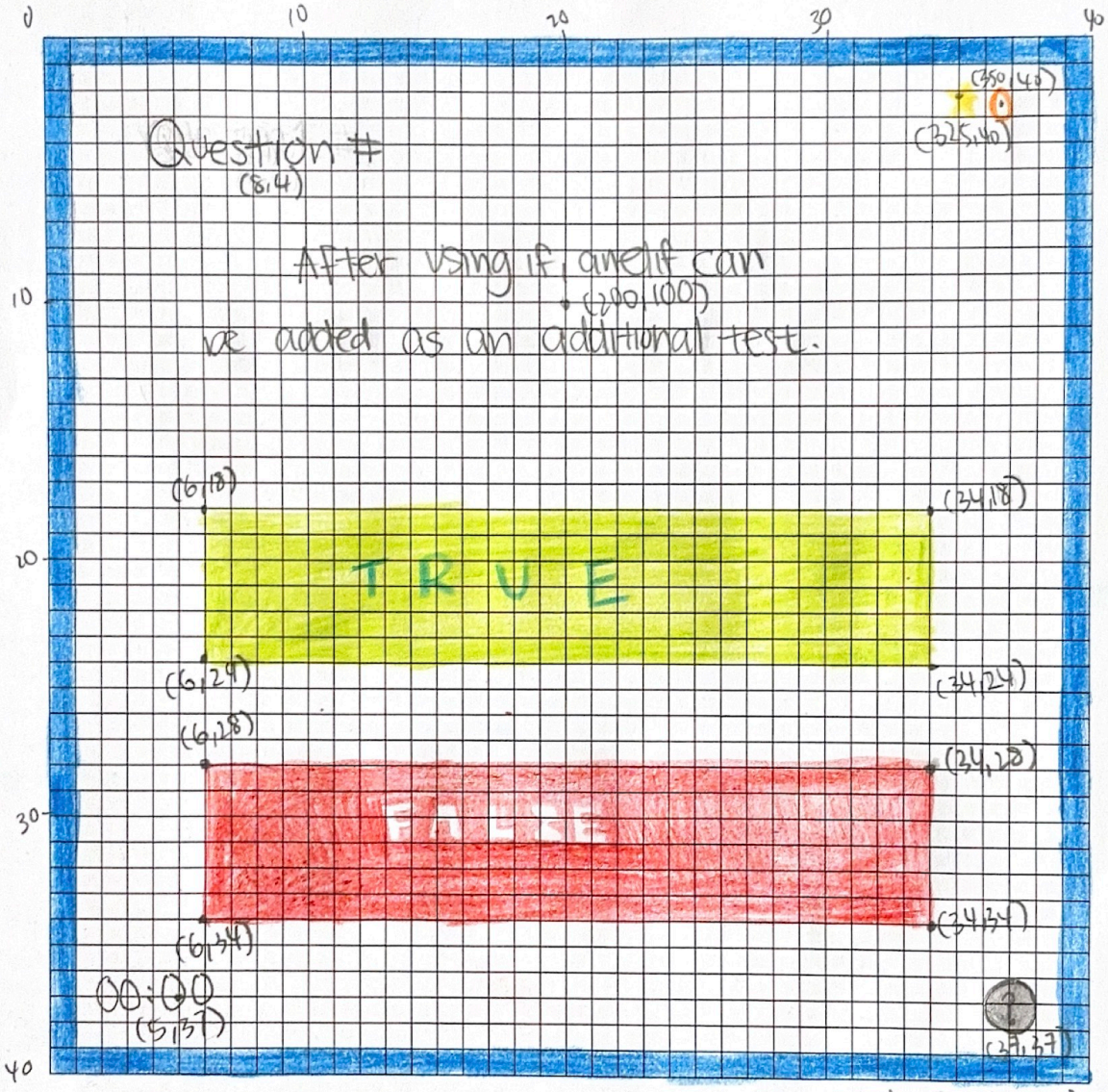
Colors

- rgb(61,220,93)
- rgb(253,188,45)
- rgb(241,80,41)
- rgb(230,23,3)
- rgb(0,122,195)
- rgb(0,41,98)
- rgb(186,189,188)
- rgb(74,79,103)
- rgb(255,255,255)
- rgb(255,225,53)

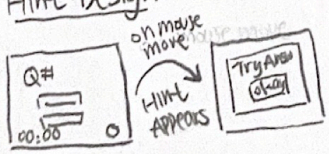
cover page:



Hint page

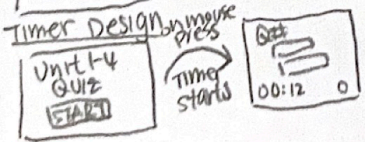


Hint Design:

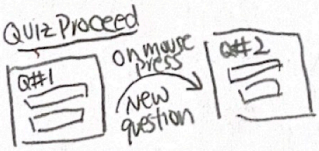


Cover Design

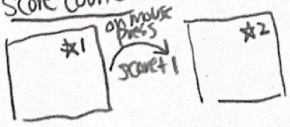
Timer Design



Quiz Proceed



Score count



Hint Design: when mouse touches ?, a hint appears. Press "okay" to unshow the hint box.

Cover Design: The Quiz starts with a cover page, press "start" to start the Quiz.

Timer Design: Timer starts after "START" is pressed.

Quiz proceed: Answer one question for the next!

Score count: with each correct answer the score increases by one

plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Variables	I will definitely need variables in my program. I will name the shapes and text created so the program is simpler to understand and code in functions. Moreover, I need variables for my if and then statements, which require variable names to function. I will also have to make sure my names are in camelCase, such as coverTitleOne and questionTitle. Global variables will also be used.
Event Functions	I plan to use 2 main functions throughout the system, which are onMouseMove(x,y) and onMousePress(x,y). For onMouseMove(x,y), it indicates that as long as the mouse touches the set object, the function is called; on the other hand, onMousePress(x,y) requires you to click the designated shape. onMouseMove(x,y) will probably be for the helper on the bottom right of the page, while onMousePress(x,y) will be designed for the rest.
if Statements	I will be using if statements throughout my program; they control for the program to change in slides after a certain condition is exceeded. I am not sure if I will be using if and else statements during the program, as the if statements will probably be similar to “hint. hits(x,y)==True”, therefore, else and elif will not be applicable. I would understand this program as a PPT, as if a true box is pressed, a whole new page is shown. In this case, the if statement checks if the box is pressed, and within it, after this, it will code for the pages to change accordingly.
Shape Objects	Space objects tend to be used in every program. I plan to use multiple drawing objects to set up my page, such as rectangles, text, circles, lines, and stars. This setup will be much simpler compared to students who aim to create visually appealing scenes, but finding the exact axis for the shapes so the page seems professional will be more challenging.
Properties	I also plan to use many properties throughout my design, such as the visibility tool and the change-value-in-a-text-box tool, to achieve appearance control and switch between questions. Properties play a key role in my event functions, as onMousePress and onMouseMove mainly involve changing the visibility of the designed text and boxes. I will use the strategy of changing the text and visibility of labels and shapes through testing with if statements; therefore, my functions may consist only of properties. This program is designed to be a Kahoot game; therefore, there will not be much changing of places or the shapes of shapes. It's just the appearance of certain words that have to be changed.
Custom Properties	Custom properties will also be something I will touch on, but not use frequently. I will be designing a scoreboard; therefore, the code must be that when the user answers the question correctly, the score.count+=1, this will probably be the only place I will use the custom property.

Shape Methods	I plan on utilising shape methods for collision detection, which triggers the move of a function. For instance, “hint. hits(x,y)==True” or “falseBox.hits(x,y)==True” which are located within the if and else functions.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

One of the most significant opportunities I found and really want to share is adding a timer at the bottom of my page. The thought was initially provided by my teacher, who noted that if I want to design a Kahoot-like system, why not add a timer, so the user knows the time they finished the quiz? I really liked this idea; however, I have not learned how to code this up to unit 4. Rather than a challenge or an aha-moment, it was more of me trying a new way to approach programming when it comes to tasks I cannot finish on my own. I asked AI for help, I told it my ideas, requests, and status on coding, so it could help me. I had to talk with AI for multiple rounds until it gave me the desired code:

```
app.time = 0
app.stepsPerSecond=1
timeLabel = Label("00:00",50,370,size=25,fill=rgb(74,79,103))

def onStep():
    if (cover.visible==False):
        app.time += 1
        minutes = app.time//60
        seconds = app.time%60

        timeLabel.value = f'{minutes:02}:{seconds:02}'
```

After this, I made sure to understand the operation instead of purely moving on after attaining the answer. I will also make sure to understand any code AI provides in the future, and only if asking AI is extremely necessary.

With this design incorporated, my code seems more realistic and reasonable, as it now has all the elements of a Kahoot and can time the user while they are answering the questions. I did not have to trade off anything to incorporate this design; I located it on the bottom left of my system, mirroring the location of the hint box.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose

Variables	80	Variables are names that store an object or certain values that can be used later in the program.	On line 80 of the program, it shows an "app.stop()" within if (score.value==3); It is a global variable, and whenever the score of the user of exceeds 3, the while program (app) will end automatically.
Event Functions	32	An event function is a function that runs when a certain goal is achieved.	onMouseMove(x,y) and onMousePress(x,y) were coded according to the plan, so I want to especially talk about my use of "onStep():", which was newly incorporated as the function that controls the timer. onStep() is an event function that runs over and over while the program is running; therefore, time is counted.
if Statements	33	An if statement indicates that the program will execute its code only if the condition is met.	I used "if (cover.visible==False)" in line 33; if this condition is met, the timer will start. This indicates that after the Start button is pressed and the cover is turned invisible, the timer starts ticking. This is a simple showcase of the use of if statements, which are just if something is met, start something.
Shape Objects	4	Shapes can be created by entering the parameters of the shape using a set form, such as for a circle, you give (centerX, centerY, radius, others).	On line 4, I created a star, and gave it the parameters, centerX, centerY, radius, points, and a color of rgb(253,188,45).
Properties	43	Properties are attributes that change the shape's appearance and location, such as setting values for visibility and center location.	Line 43 states that "hintBox.visible=True", which means that when the condition in the if statement is met, the visibility of the hintBox will turn true; otherwise, it stays false. The visibility of the hint box is a property.
Custom Properties	62	Custom properties are properties created by the programmer to store data inside an object.	On line 62, I programmed the value of the score to increase by one if the condition of the if statement is true. The "score" is not a property within Java itself; it is something I created and named, ascribing special meaning to it.

Shape Methods	61	Shape methods are built-in functions that check on shape objects and control the spe to perform actions, like “shape.hits(x, y)” or “toFront()”.	Line 61 is a type of shape method “trueBox.hits(x,y)==True”, which means that if the mouse and the true box hit, the code following the if statement will be performed. trueBox.hits(x,y) is the shape method in this case, it checks whether the trueBox and the mouse collide.
Arithmetic Operators	35 - 36	Arithmetic operators are used for calculations in the program. Common operators used include =, +=,==,+, //, and %.	I want to especially point out the use of // and % on lines 35 and 36, which are provided by AI. // indicates a division followed by a floor to create an integer. On the other hand, % means Modulus, which is the remainder after a division. These 2 concepts combined give me the exact time.
Formatted Strings (f-strings)	38	Format this string using variables inside { }	Formatted Strings were something AI provided me with in the code, and they control how the time will be shown in the form of {minutes:02}:{seconds:02} within formatted strings.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

If I were to redo the coding process, I would absolutely add the complexity of the game. I wanted to add more questions; however, due to time concernes, and that the coding will be rather similar, I did not side with adding more questions, instead, I focused on adding more elements, such as the hint box or the timer, to show my full understanding of the course. However, if I had more time, I would “finish” the game by adding around 7 questions.

Moreover, adding a timer was a significant adjustment in my code; I increased the complexity of the process while not posing contradictions with the other programs. This process was a complication; however, with the help of AI, the process went much smoother and ideal. The number of questions also decreased from 10 to 3, because creating a new question took much more time and energy than I thought. I wouldn't say this is beneficial, but it does save me time that I can spend on other elements. If I have another chance to redo the planning process, I should have thought of it over more thoroughly and planned something more reasonable, such as planning about 5 questions instead of 10.

Resource Utilization is absolutely something I would like to point out. I used the “cheat sheet” provided by CMU CS Academy, and it reduced a lot of my research time. I also looked over the classes of CMU CS Academy, including the debugging, creative tasks, and lesson reviews, where I found inspiration. My teacher also checked on my progress and gave feedback, such as adding questions randomly (which I did not add) or the timer idea. Lastly, peers also gave feedback, they tried the program, and suggested a tweak on the aesthetic, which I changed accordingly.

If I had a chance to redo the process, I would definitely start early, having 10 questions done, and a higher complexity on this project isn't a hard thing; it is just a matter of time. With the idea already formed in my head, I should have started during the long break, when I had abundant time to code for what I wanted. However, despite being late, I was still able to produce a fully developed system where I chose to incorporate a reasonable number of elements in the limited time I had. Overall, I effectively used all my time, but it could have been better if I had started earlier.

Personally speaking, I feel this project was less complete than the first Unit 2 project. I would ascribe this to the limited time (especially since I have one fewer class than my classmates) and the limited number of elements my idea needed. A Kahoot game has nothing beyond the score, time, cover page, hint, and questions, so I feel that even if I wanted to add anything, it wouldn’t be something I could finish as a beginner in Java.

Scroll down for submission checklist.

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Pre-Submission Checklist for Creative Tasks

Program Design

- Described what your program will do clearly.
- Incorporated at least two concepts from the current unit.
- Added annotations to explain colors, movements, and user interactions.

Canvas Design

- Sketched your canvas layout.
- Labeled any animation or user interaction on the canvas sketch.
- Detailed dominant colors and their importance.
- Described any animations or movements.

Code Design

- Listed at least four to six coding concepts or properties you plan to use.
- Explained why each concept or property is essential for your program.

Program Development and Coding

- Finished coding the program.
- Debugged and tested the program.
- Checked if the program meets the design criteria.

Difficulties and Opportunities

- Described a difficulty faced or an opportunity realized in detail.
- Explained the steps taken to resolve the difficulty or seize the opportunity.

Finished Project Code Outline

- Listed only one instance of each important coding concept used.
- Provided the line numbers for each coding concept.
- Included a brief definition for each coding concept.
- Highlighted the aspect you are most proud of.

Future Goals

- Discussed what specific changes you'd make if given more time.
- Reflected on your approach and planning for the project.
- Considered resource utilization and time management.

Final Check

- Revisited the rubric to make sure you meet all the criteria.
- Made sure your document is well-organized and free of grammatical errors.
- Ready to submit the project and this document.

By ticking off each item on this checklist, you ensure that you have met all the requirements and can confidently submit your work for evaluation.